

Q1. Given a 2D array, print all elements row by row

Input	Output
<pre>matrix = [[1, 2, 3], [4, 5, 6], [7, 8, 9]]</pre>	<pre>1 2 3 4 5 6 7 8 9</pre>

Q2. Print the matrix column by column

Input	Output
<pre>matrix = [[1, 2, 3], [4, 5, 6], [7, 8, 9]]</pre>	<pre>1 4 7 2 5 8 3 6 9</pre>

Q3. Find the total sum of all elements in the matrix.

Input	Output
<pre>matrix = [[1, 2, 3], [4, 5, 6]]</pre>	<pre>21</pre>

Q4. Return the maximum and minimum value from the matrix

Input	Output
<pre>matrix = [[10, 5, 8], [3, 20, 7]]</pre>	<pre>Maximum = 20 Minimum = 3</pre>

Q5. Count how many even and odd numbers are in the matrix.

Input	Output
<pre>matrix = [[1, 2, 3], [4, 5, 6]]</pre>	<pre>Even = 3 Odd = 3</pre>

Q6. Find the sum of each row and return a 1D array.

Input	Output
<pre>matrix = [[1, 2, 3], [4, 5, 6], [7, 8, 9]]</pre>	<pre>[6, 15, 24]</pre>

Q7. Find the sum of each column and return a 1D array.

Input	Output
<pre>matrix = [[1, 2, 3], [4, 5, 6], [7, 8, 9]]</pre>	<pre>[12, 15, 18]</pre>

Q8. Return the row index that has the maximum sum.

Input	Output
<pre>matrix = [[1, 2, 3], [10, 20, 30], [4, 5, 6]]</pre>	<pre>Row Index = 1 Maximum Row Sum = 60</pre>

Q9. Return the column index that has the maximum sum.

Input	Output
<pre>matrix = [[1, 20, 3], [4, 50, 6], [7, 8, 9]]</pre>	Column Index = 1 Maximum Column Sum = 78

Q10. Find the sum of the primary diagonal.

Input	Output
<pre>matrix = [[1, 2, 3], [4, 5, 6], [7, 8, 9]]</pre>	15

Q11. Find the sum of the secondary diagonal

Input	Output
<pre>matrix = [[1, 2, 3], [4, 5, 6], [7, 8, 9]]</pre>	15

Q12. Find the absolute difference between primary diagonal sum and secondary diagonal sum

Input	Output	
<pre>matrix = [[11, 2, 4], [4, 5, 6], [10, 8, -12]]</pre>	15	Primary diagonal = $11 + 5$ $+ (-12) = 4$ Secondary diagonal = $4 +$ $5 + 10 = 19$ Difference = $4 - 19 =$ 15

Q13. Find the sum of elements above the primary diagonal

Input	Output
<pre>matrix = [[1, 2, 3], [4, 5, 6], [7, 8, 9]]</pre>	11

Q14. Find the sum of elements below the primary diagonal.

Input	Output
<pre>matrix = [[1, 2, 3], [4, 5, 6], [7, 8, 9]]</pre>	19

Q15. Return the transpose of the matrix

Input	Output
<pre>matrix = [[1, 2, 3], [4, 5, 6]]</pre>	<pre>[[1, 4], [2, 5], [3, 6]]</pre>

Q16. Given two row indexes, swap those rows.

Input	Output
<pre>matrix = [[1, 2, 3], [4, 5, 6], [7, 8, 9]] row1 = 0 row2 = 2</pre>	<pre>[[7, 8, 9], [4, 5, 6], [1, 2, 3]]</pre>

Q17. Given two column indexes, swap those columns.

Input	Output
<pre>matrix = [[1, 2, 3], [4, 5, 6], [7, 8, 9]] col1 = 0 col2 = 2</pre>	<pre>[[3, 2, 1], [6, 5, 4], [9, 8, 7]]</pre>

Q18. Rotate a square matrix 90 degrees clockwise.

Input	Output
<pre>matrix = [[1, 2, 3], [4, 5, 6], [7, 8, 9]]</pre>	<pre>[[7, 4, 1], [8, 5, 2], [9, 6, 3]]</pre>

Q19. Rotate a square matrix 90 degrees counterclockwise.

Input	Output
<pre>matrix = [[1, 2, 3], [4, 5, 6], [7, 8, 9]]</pre>	<pre>[[3, 6, 9], [2, 5, 8], [1, 4, 7]]</pre>

Q20. Find the sum of each column. Then return the difference between consecutive column sums.

Input	Output	
matrix = [[1, 3, 1], [6, 4, 2], [5, 1, 7], [9, 3, 3], [8, 5, 4]]	[-13, 1]	Column 0 = $1 + 6 + 5 + 9 + 8 = 29$ Column 1 = $3 + 4 + 1 + 3 + 5 = 16$ Column 2 = $1 + 2 + 7 + 3 + 4 = 17$ $16 - 29 = -13$ $17 - 16 = 1$

Q21. Find the row with the smallest sum and the column with the smallest sum. Return the value at their intersection.

Input	Output	
matrix = [[50, 45, 90, 35], [70, 80, 50, 95], [40, 30, 25, 75]]	25	Explanation: Smallest row sum is in row 2. Smallest column sum is in column 2. Intersection value = $\text{matrix}[2][2] = 25$.

Q22. Find the row with the maximum sum and the column with the maximum sum. Return the value at their intersection.

Input	Output	
matrix = [[3, 3, 6], [7, 4, 9], [1, 2, 8]]	9	Explanation: Largest column sum is in column 2. Largest row sum is in row 1.

		Intersection value = matrix[1][2] = 9
--	--	---

Q23. Each row represents a student. Each column represents a quiz. Find the average score of each quiz. Then return the difference between consecutive quiz averages.

Input	Output	
matrix = [[85, 90, 78], [88, 85, 80], [75, 95, 85], [92, 88, 82]]	[4.5, -8.25]	Quiz 1 average = $(85 + 88 + 75 + 92) / 4 = 85.0$ Quiz 2 average = $(90 + 85 + 95 + 88) / 4 = 89.5$ Quiz 3 average = $(78 + 80 + 85 + 82) / 4 = 81.25$ Quiz 2 - Quiz 1 = $89.5 - 85.0 = 4.5$ Quiz 3 - Quiz 2 = $81.25 - 89.5 = -8.25$

Q24. Each row represents a month. Each column represents a product. Find the total sales of each product and return the percentage difference between consecutive products.

Input	Output	
matrix = [[100, 200, 300], [150, 250, 350], [120, 180, 290]]	[13.40, 15.98]	Product totals: Product 0 = $100 + 150 + 120 = 370$ Product 1 = $200 + 250 + 180 = 630$ Product 2 = $300 + 350 + 290 = 940$ Total sales: $370 + 630 + 940 = 1940$

		Contribution: Product 0 = $370 / 1940 * 100 = 19.07\%$ Product 1 = $630 / 1940 * 100 = 32.47\%$ Product 2 = $940 / 1940 * 100 = 48.45\%$ Difference: Product 1 - Product 0 = $32.47 - 19.07 = 13.40$ Product 2 - Product 1 = $48.45 - 32.47 = 15.98$
--	--	--

Q25. Find a cell that is smaller than all of its valid neighboring cells: top, bottom, left, and right. Return the position of the local minima. If multiple local minima exist, return the first one found from row-wise traversal.

Input	Output	
<pre>matrix = [[21, 21, 1000, 23], [25, 17, 21, 1000], [1000, 19, 18, 22], [21, 22, 1000, 20]]</pre>	[2, 2]	Value at position [2, 2] is 18. Its valid neighbors are: Top = 21 Bottom = 1000 Left = 19 Right = 22 Since 18 is smaller than all of them, [2, 2] is a local minima.

Prepared and Organized by
BADHON NANDI
Student Tutor & FYAT Mentor
Department of Computer Science and Engineering, Brac University

Editing in progress

Email me for any question additions/enquiries: badhon.nandi@g.bracu.ac.bd